

Redacted Resume for NDSEG Panel Evaluation

NDSEG Applicant Summary

PhD student in Mechanical Engineering specializing in scientific machine learning, with a strong track record of conducting and communicating high-impact research. Skilled at translating complex scientific concepts into clear insights, combining a deep understanding of mechanical engineering with advanced proficiency in Python and machine learning. First-author on multiple published research papers and experienced in presenting work at national and international conferences. Recipient of multiple awards and fellowships supporting doctoral research, demonstrating recognized excellence in both technical expertise and scholarly contribution.

Education & Certifications

- 2022, Bachelor of Science, Mechanical Engineering, GPA: 3.97
- 2024, Master of Science, Mechanical Engineering and Computer Engineering, GPA: 4.0

Skills

- Problem-solving
- Leadership
- Time management
- Independent learning
- Python
- Matlab
- C++
- JavaScript
- Git
- PyTorch
- Scientific machine learning

Awards

- Scholarship awarded by a large research conference recognizing my academic and personal accomplishments
- Fellowship awarded by my Ph.D. institution for the first two years of my research recognizing my academic accomplishments
- Additional scholarship awarded by my Ph.D. institution further recognizing outstanding academic performance
- Research Excellence Award awarded by my Master of Science institution recognizing my outstanding research performance during my master's degree
- Graduated Summa Cum Laude for both bachelor's and master's degrees
- Additional scholarship awarded by a federal institution recognizing academic performance
- Award given by my undergraduate institution recognizing my outstanding performance as a student organization leader for that school year

Publications

- Automated Camera Calibration (ACC): A Python GUI Application for Automatically Calculating Camera Intrinsics with Arri Camera Systems – Published in a peer-reviewed journal for open-source software
- Two Automated Point Cloud Filtration Approaches using Principal Component Variance and Nearest Neighbor Distance - Published in a peer-reviewed journal for imaging technology
- Automated Point Cloud Filtration Through Minimization of Point Cloud Metrics – Presented at a conference for imaging technology
- Streamlining Point Cloud Post-Processing Using Principal Component Variance, Distribution Evaluation, and Other Statistical Metrics - Presented at a conference on technology applicable to the Department of War
- A Flexible Multi-Modal Multi-User Traffic Simulation for Studying Complex Road Design – Published in the proceedings of a conference on computers in engineering

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Experience	
Position Title: Graduate Research Assistant	Dates: 08/2024 - Present
- Conduct graduate-level research applying machine learning methods to materials engineering challenges in a collaborative team environment - Developed an independent research project on neural operators, from problem formulation to model implementation and evaluation - Apply Python-based machine learning frameworks to analyze, model, and interpret complex materials data. - Coordinate with interdisciplinary researchers to integrate mechanical engineering techniques with state-of-the-art computational methods	
Position Title: Graduate Research Intern	Dates: 05/2023 - Present
- Support machine learning research team by developing and maintaining virtual environments for experimentation and deployment - Teach colleagues how to implement and apply specific machine learning techniques, improving team efficiency - Contribute to independent development projects, applying coding and research skills to explore new methods and tools - Competed in and won organization-wide machine learning contest by creating a highly accurate conditional generative adversarial network	
Position Title: Graduate Research Assistant	Dates: 01/2022 – 05/2024
- Conducted graduate research resulting in multiple journal and conference publications in mechanical engineering and machine learning - Designed and developed a virtual reality demonstration to showcase research applications during lab tours and outreach events - Led engaging lab tours for groups of elementary and middle school students, communicating complex engineering concepts in accessible ways	
Position Title: Additive Manufacturing Intern	Dates: 05/2021 – 12/2021
- Operated and maintained both metal and plastic 3D printers, ensuring reliable performance and high-quality output - Managed printer schedules to maximize efficiency and meet project deadlines - Diagnosed and resolved mechanical and software issues to minimize downtime and maintain productivity - Advised colleagues on part design strategies to improve printability and optimize manufacturing outcomes	
Position Title: Undergraduate Research Assistant	Dates: 01/2019 – 12/2021
- Assisted in research projects by supporting data analysis, literature review, and experimental design tasks - Contributed to the writing and preparation of a published conference paper - Collaborated with graduate students and faculty to advance project goals and meet research deadlines - Gained foundational experience in conducting academic research and communicating scientific findings	
Position Title: Simulation Intern	Dates: 05/2020 – 08/2020
- Assisted in developing Python scripts to identify harmonic frequencies in engines using Fourier transform techniques - Applied lumped dynamic modeling to analyze engine vibration behavior and predict resonance conditions	